Networking is the process of connecting computers and devices so they can communicate and share data with each other.

Real-Time Example

Imagine a company

- 50 computers 🖥️
- 2 printers 🖨️
- Internet connection 🌐
- CCTV cameras 📹
- Servers 🗄️

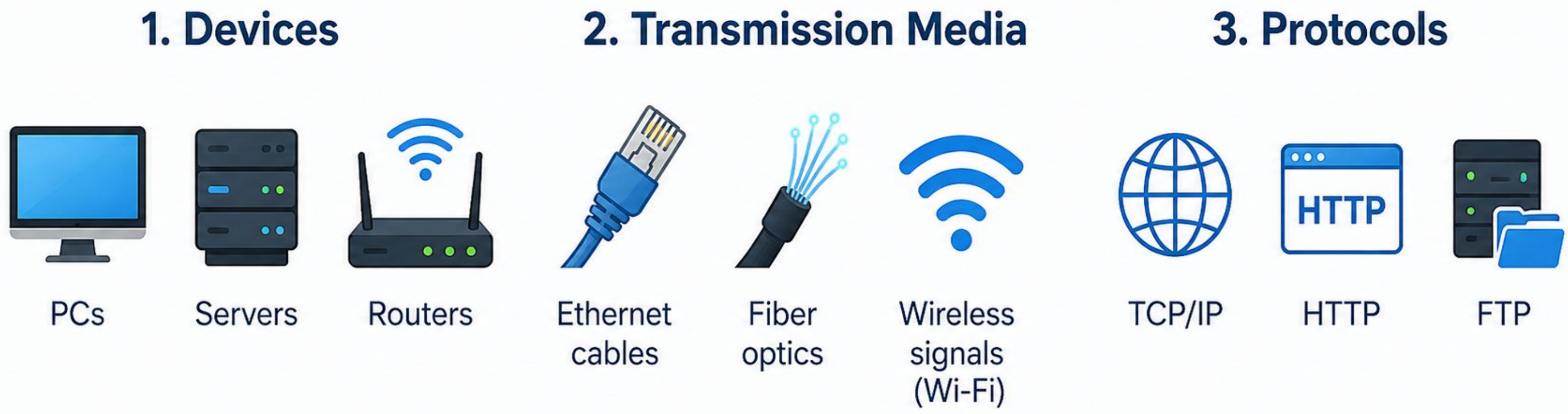
Without networking:

- No file sharing ❌📂
- No internet sharing ❌🌐
- No centralized management ❌⚙️
- USB drives running around 💾

With networking:

- Access resources easily ✅📂
- Communication becomes faster ⚡💬
- Data sharing centralized 🔄🗄️

Main Components in Networking



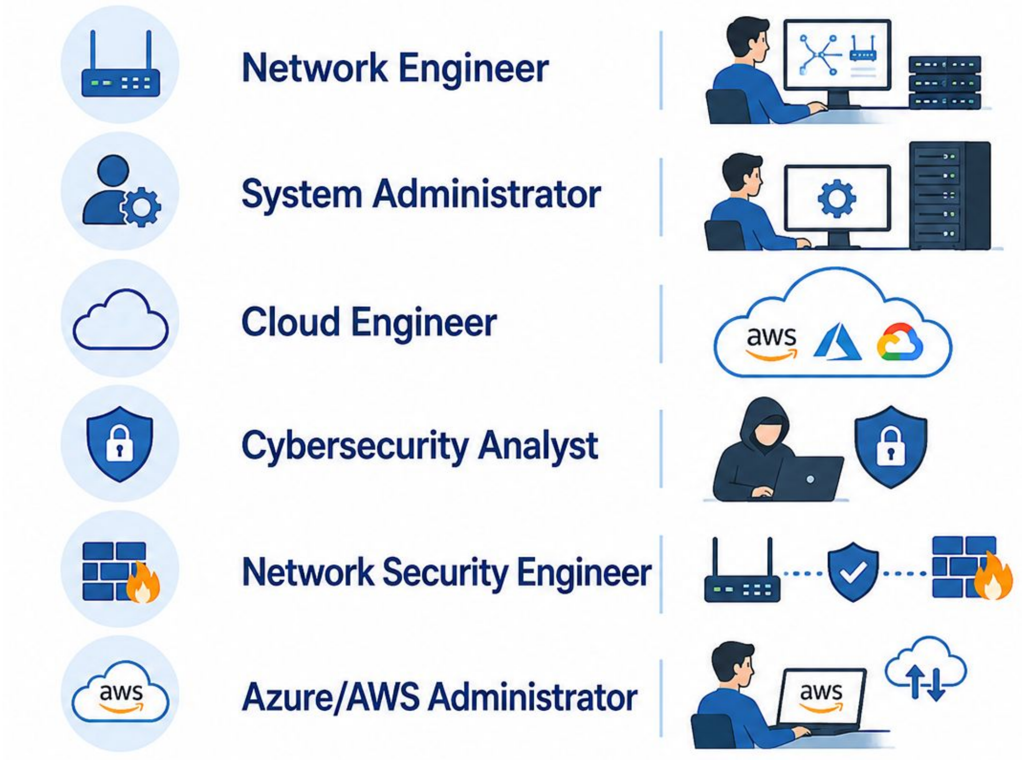
Types of Networks

Type	Meaning	Example
LAN	Local Area Network	Office network
WAN	Wide Area Network	Internet
MAN	Metropolitan Area Network	City-wide network
WLAN	Wireless LAN	Wi-Fi network

Where Network is Important

- Internet access
- Cloud computing
- Online banking
- Video calls
- Cybersecurity
- Data centers
- Azure/AWS/Cloud infrastructure

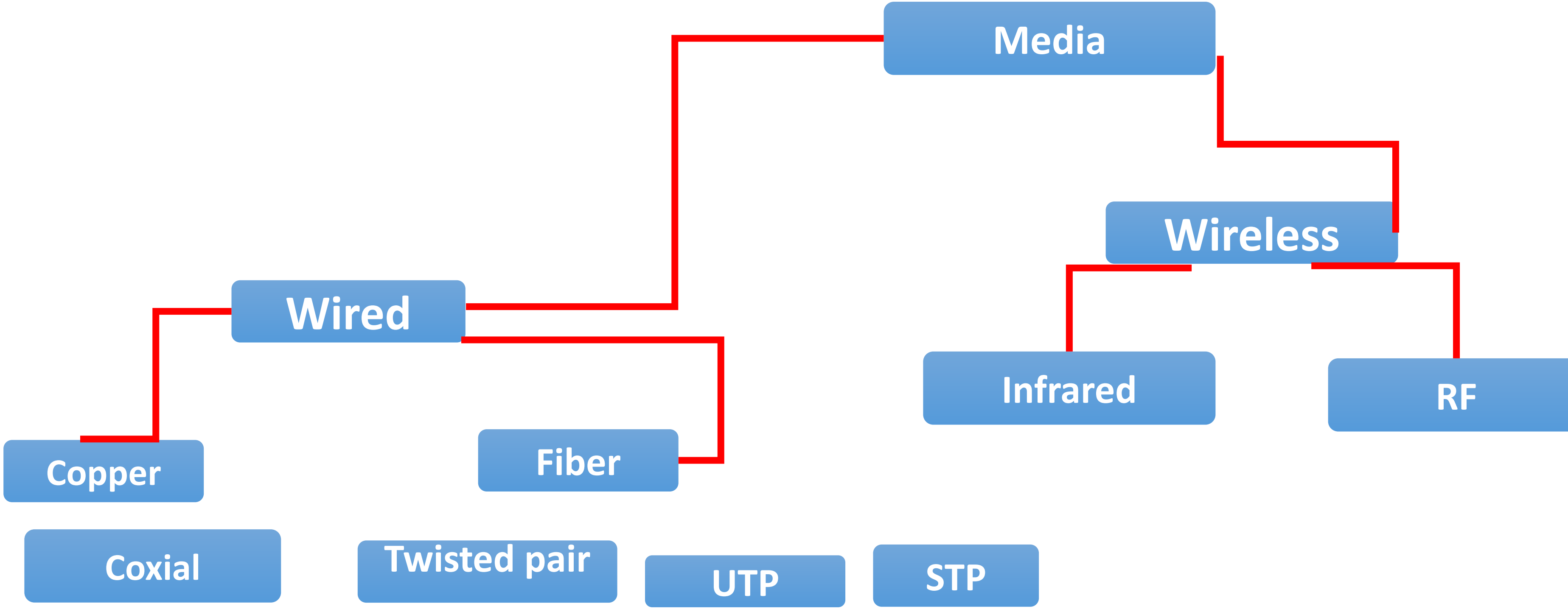
Career Opportunities in Networking



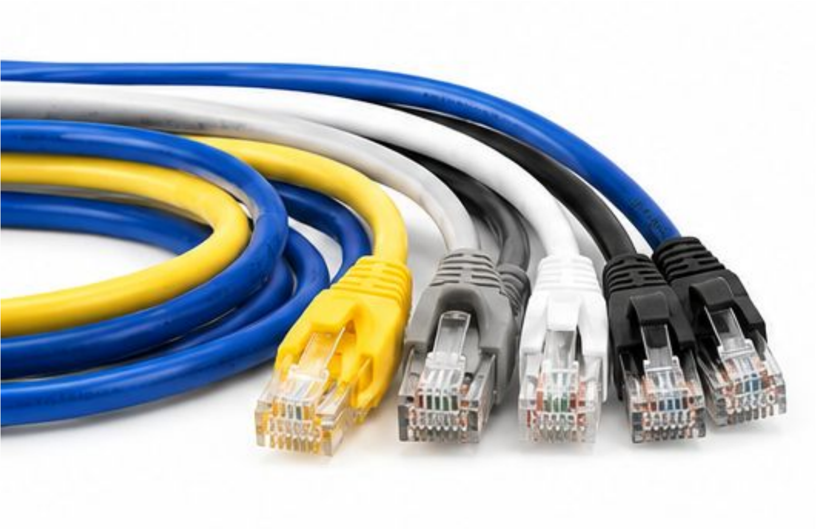
Popular Networking Certifications

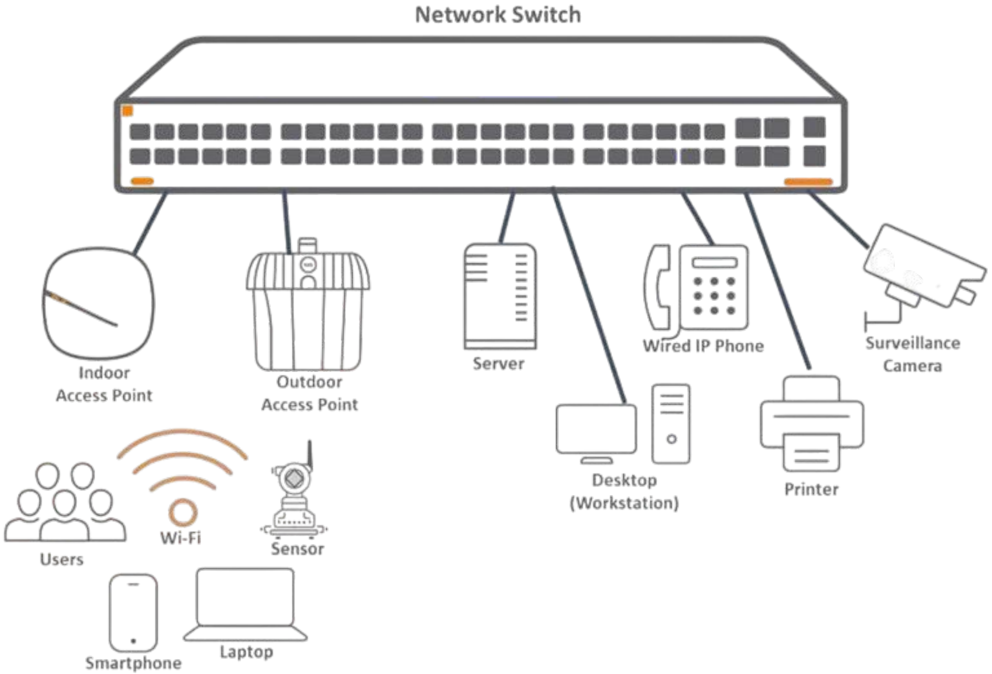
Media

- The purpose of the media is to transport bits from one machine to another.



UTP cables




Network Switch
Indoor
Access Point
Outdoor
Access Point
Server
Wired IP Phone
Surveillance
Camera
Desktop
(Workstation)
Printer
Users
Wi-Fi
Sensor
Smartphone
Laptop

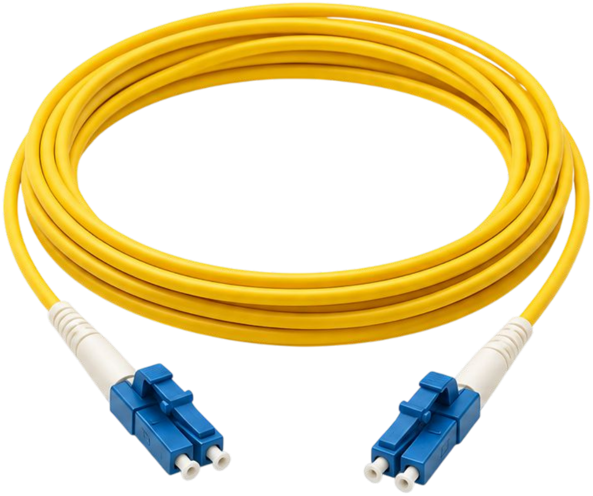
Types of Twisted Pair Cables

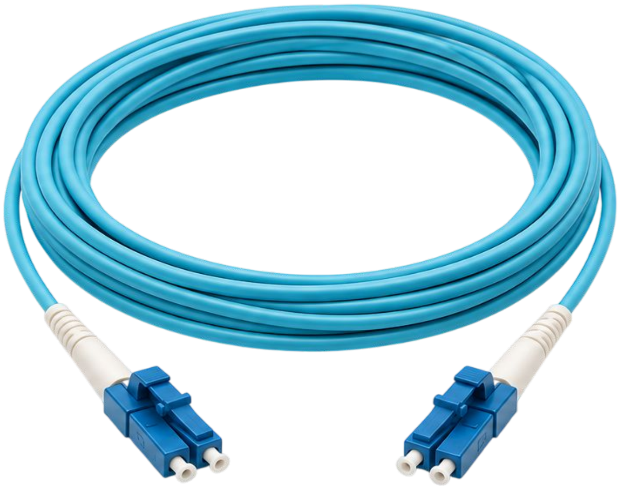
Category	DTR (Data Transfer Rate)	Purpose	Connector
CAT 1	1 Mbps	Telephone Lines	RJ 11
CAT 2	4 Mbps	Telephone Lines	RJ 11
CAT 3	10 Mbps	Ethernet	RJ 45
CAT 4	16 Mbps	Ethernet	RJ 45
CAT 5	100 Mbps	Fast Ethernet	RJ 45
CAT 5e	500 Mbps	Fast Ethernet	RJ 45
CAT 6	1000 Mbps	Gigabit Ethernet	RJ 45

FIBER CABLES

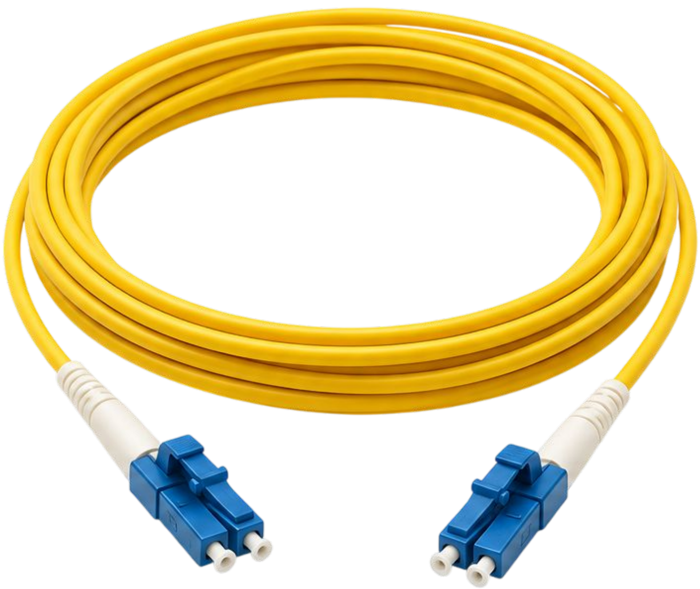
1.Single Mode Fiber

2.Multi Mode Fiber



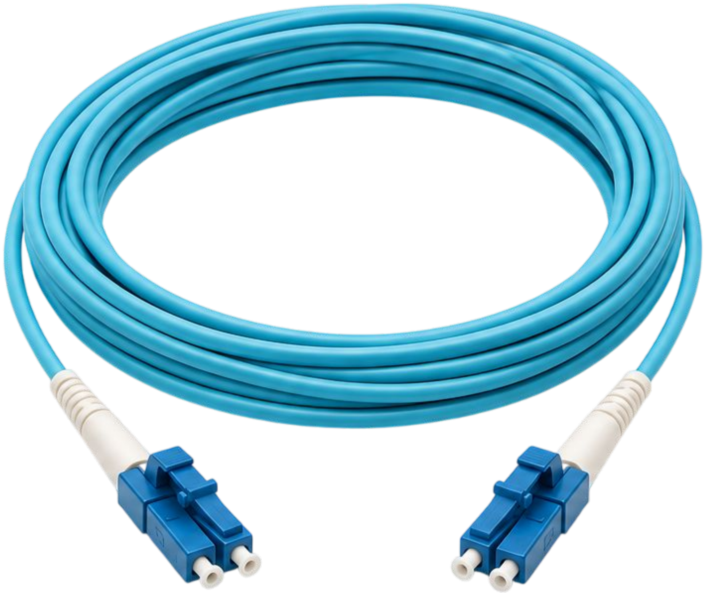


1.Single Mode Fiber



1. **Core Size: 8–10 µm**
   Small core allows light to travel in a single path.
2. **Light Source: Laser**
   Uses laser light for high-speed, long-distance transmission.
3. **Distance: Up to 100+ km**
   Supports communication over very long distances.
4. **Speed: Very High**
   Provides high bandwidth and very fast data transfer.
5. **Signal Loss: Very Low**
   Minimal signal loss ensures reliable performance.

2.Multi Mode Fiber

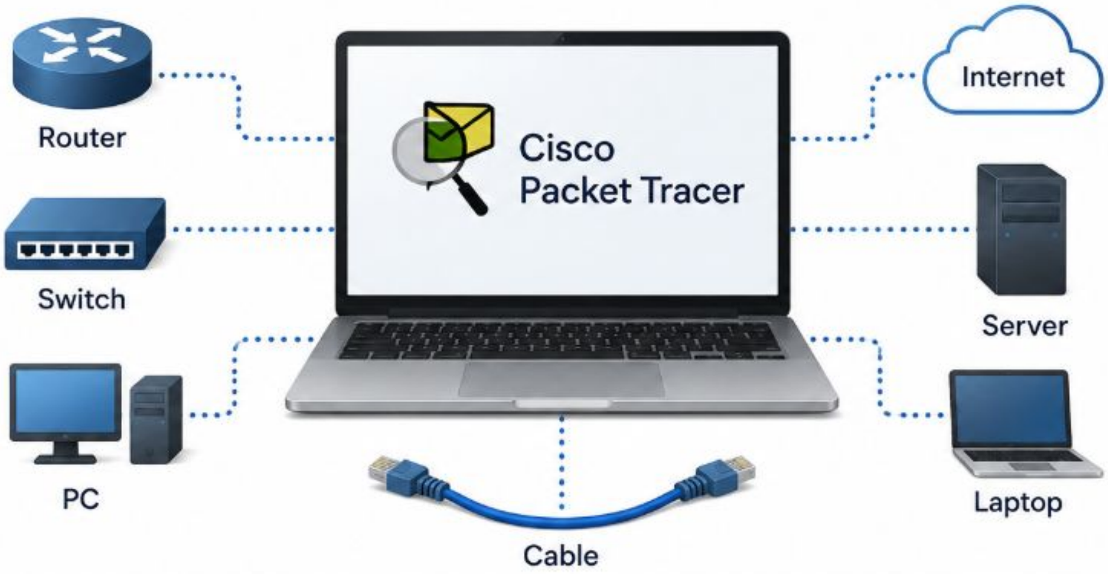


1. **Core Size: 50 / 62.5 μm**
   Larger core allows multiple light modes (paths) to travel.
2. **Light Source: LED**
   Uses LED light for shorter distance communication.
3. **Distance: Up to 2 km**
   Ideal for short-distance applications like LANs and data centers.
4. **Speed: Moderate (10G/40G/100G)**
   Supports high-speed connections over short distances.
5. **Signal Loss: Higher**
   Higher attenuation compared to single-mode fiber.

What is a Packet Tracer?

Cisco Packet Tracer is a network simulation software made by **Cisco Systems**.

It lets you create virtual networks on your computer instead of buying actual routers, switches, and cables like a small ISP.



SWITCH

- Connects Multiple PCs
- Maintains MAC Address Table
- Used in Local Area Network (LAN)